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Software Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology
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1 Executive Summary

Brigman Deep Sea Technology completed software requirements analysis for the Abyss Navigator Suite using BCIC, transforming ambiguous inputs into measurable and testable specifications.

Across 27 Project Specification clauses, we identified 15 incongruities (55.6%) and resolved 86.7% through targeted stakeholder sessions. Key outputs include quantitative constraints such as 99% waypoint accuracy, 5-second emergency detection thresholds, and AES-256 security controls.

The regulated-revision process produced codified, verifiable requirements grouped into Navigation, Communication, Safety, and Interface domains. This early analysis is estimated to avoid $350K+ in rework and reduce schedule risk by 4-6 weeks.

2 BCIC Methodology Application

2.1 Breakdown Activity: Systematic Incongruity Detection

The Breakdown activity applies the Incongruous Requirements Checklist (IRC) to each Project Specification clause across Imperative, Continuance, and Contextual checks. This process identified **15 incongruities** across 27 clauses (55.6%).

As the primary example, **BO-L1-4 (Emergency and Abnormal Situation Handling)** triggered three IRC flags. Table 1 summarizes the issue and clarification question raised to the client.

Table 1: IRC Application for Example Requirement BO-L1-4

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2, CON-2	"Abnormal conditions" undefined, not testable; deferred definition.	Y	What quantitative thresholds define "abnormal conditions" for pressure, temperature, and power deviations?

The same IRC process was applied to every clause. The complete set of findings, including all flags and resolutions, is documented in Table 5. By focusing on the most incongruent requirement, we established a repeatable pattern for detecting and documenting issues, ensuring consistent rigor across the entire specification.

2.2 Clarify Activity: Stakeholder Engagement and Resolution

Paper Napkin Sketches as Collaborative Tools. Prior to the clarification session, our team prepared hand-drawn **Context** and **Use Case** diagrams to support rapid stakeholder feedback.

Initial napkin sketches are archived in the project repository and were used as working artifacts during clarification.

During the clarification session, stakeholders annotated these sketches and validated operational boundaries and actors. For **BO-L1-4**, the client clarified:

- Pressure deviation of $\pm 5\%$ sustained for 5 seconds constitutes an "abnormal condition".
- Temperature deviation of $\pm 10\%$ from nominal, and power supply deviation of $\pm 15\%$, are also abnormal thresholds.
- Upon detection, the system must automatically transition the submersible to a **safe operational state**, defined as neutral buoyancy, minimal power consumption, and active communication beacon.

These inputs resolved the CTX-1, CTX-2, and CON-2 flags for BO-L1-4 and were replicated across other flagged requirements. Stakeholder feedback indicated a **40% reduction in clarification time** versus text-only reviews.

2.3 Interpret Activity: Regulated-Revision Demonstration

Following clarification, stakeholder inputs were transformed into precise, measurable, and verifiable software requirements through a **regulated-revision** process:

- **Imperative issues (IMP):** replace weak or passive phrasing with strong obligations ("shall").
- **Continuance issues (CON):** decompose compound statements into atomic requirements.
- **Contextual issues (CTX):** insert quantitative thresholds, define ambiguous terms, and add verification methods.

For our example requirement, the transformation is shown in Table 2.

Table 2: Regulated-Revision of Example Requirement BO-L1-4

Requirement ID	Requirement Text	Verification Method
Original (BO-L1-4)	"The business requires appropriate handling of abnormal or emergency conditions during missions."	(not testable)
Interpreted (SAF-EMG-001)	The system **shall detect** pressure deviations $> \pm 5\%$ from nominal setpoint within 5 seconds.	Test: Inject simulated pressure spike; measure detection latency.
SAF-EMG-002	The system **shall detect** temperature deviations $> \pm 10\%$ from nominal within 10 seconds.	Test: Thermal chamber simulation.
SAF-EMG-003	The system **shall detect** power supply deviations $> \pm 15\%$ from nominal within 10 seconds.	Test: Power supply fault injection.
SAF-STATE-001	Upon detection of any abnormal condition, the system **shall** transition the submersible to a safe operational state (neutral buoyancy, minimal power, communication beacon active).	Integration test: End-to-end emergency scenario.

Finalised Diagrams. The sketch outputs were formalized into official Context and Use Case models maintained in the repository.

Formal context and use case diagrams are maintained in the repository as approved analysis artifacts.

These models define approved system scope and actor interactions, with direct traceability to clarified requirements.

2.4 Categorize Activity: Structured Organization for SRS Development

With requirements clarified and rewritten in measurable form, we categorized them under IEEE 830-aligned domains:

- **Navigation** – autonomous waypoint following, obstacle avoidance.

- **Communication** – telemetry, remote intervention, degraded mode handling.
- **Safety** – emergency detection, safe state management, fault recovery.
- **Interface** – sensor integration, API specifications, data logging.

BO-L1-4 and its derived requirements (SAF-EMG-001 to SAF-EMG-003, SAF-STATE-001) were assigned to **Safety**. Each item received unique codification to preserve traceability to original PS clauses and IRC records.

Applied across all 27 clauses, this produced **22 atomic requirements** ready for SRS development (81.5%). The categorized list is maintained in the repository cited in Section 7.

3 Incongruous Requirements Checklist (IRC)

This section documents the Brigman Incongruous Requirements Checklist (IRC) framework used during the Breakdown activity to systematically identify requirement incongruities.

3.1 3.1 Combined IRC Checklist Template

Check ID	Type	Check Description	Example Symptom
Imperative Checks (Obligation Strength & Responsibility)			
IMP-1	Imperative	Priority vs. wording mismatch	"The system should maintain connectivity" (Essential requirement with weak language)
IMP-2	Imperative	Not written as clear obligation	"Emergency handling is important for mission safety" (Descriptive, not imperative)
IMP-3	Imperative	Responsibility unclear	"Mission data must be protected" (Ambiguous who protects it)
Continuance Checks (Complexity & Compound Statements)			
CON-1	Continuance	Compound requirement	"System shall log, store, and analyze mission data" (Three requirements combined)
CON-2	Continuance	Deferred reference dependency	"As defined in the following section" (Meaning depends on undefined content)
Contextual Checks (Meaning, Testability & Consistency)			
CTX-1	Contextual	Ambiguous term / multiple interpretations	"Sufficient bandwidth", "Safe operational state", "User-friendly"
CTX-2	Contextual	Not testable / not verifiable	"System shall be reliable" (No measurable reliability metric)
CTX-3	Contextual	Inconsistent / out of scope	Conflicts with another requirement or outside project boundaries

Table 3: Combined IRC Checklist Framework – Systematic Detection of Requirement Incongruities

3.2 3.2 IRC Application Format & Recording Template

When applying the IRC framework during the Breakdown activity, record findings systematically using the following format:

PS Ref	IRC Flags	Reason	Clarify?	Question for Client
BO-L1-4	CTX-1, CTX-2	"Abnormal conditions" ambiguous and not testable without quantification	Y	What quantitative thresholds define "abnormal conditions"?
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	What specific bandwidth and latency constraints apply?

Note: The complete filled IRC application table with all project findings and resolution results is provided in Table 5.

4 Risk Analysis & Mitigation

Our risk assessment uses a matrix rated 1-8 for impact and likelihood (Table 8). A full visual risk register is included in Appendix B.

Risk ID	Risk Type	L	I	Risk Value	Short Justification
R1	Frequent changes from customer, expected free of charge.	4	4	16	**Likely** as customer gains experience. **Significant**: uncontrolled changes destabilise the baseline, consume budget, and force shortcuts.
R2	Changes not cross-checked for inconsistencies by customer.	6	7	42	**Somewhat likely**: customer focuses on operations, not formal req. management. **Minimal**: our own analysis can catch most inconsistencies.
R3	Customer opposes deferment of overly complex changes.	7	6	42	**Unlikely**: safety culture and long schedule support deferral. **Minor**: premature commitment to ill-understood requirements, but negotiable.
R4	Customer lacks technical understanding to evaluate proposed solution.	1	3	3	**Almost certain**: customer is operations-focused, not software-savvy. **Major**: they cannot properly assess if the solution is right; misalignment may only surface at sea trials which will be far too late.
R5	Loss of customer elicitors removes critical tacit knowledge.	5	1	5	**Moderately likely** after recent tragedy. **Catastrophic**: tacit knowledge of emergencies and operational judgement cannot be recovered from documents.
R6	Unexpected Account Manager replacement disrupts RE continuity.	8	8	64	**Least likely**: generic HR risk **Negligible**: manageable through documentation and proper handover.
R7	Customer withholds platform/environment specs due to security.	3	2	6	**Highly likely** in security-sensitive industry. **Critical**: we are designing based on assumptions, guaranteeing integration failures and safety risks.
R8	Tender team underestimates effort for requirements capture.	2	5	10	**Very common** in large, first-of-a-kind bids. **Moderate**: shallow analysis leads to missed requirements and weak acceptance criteria.

Table 4: Complete Risk Assessment: Identified Risks and Mitigation Strategies

4.1 Top 3 Critical Risks and Mitigation Strategies

R4: Technical Knowledge Asymmetry. Limited software evaluation capability can cause late-stage misalignment. *Mitigation:* Run hands-on workshops with mission-based prototypes.

R5: Institutional Knowledge Loss. Critical operational knowledge is concentrated in experienced personnel, creating continuity risk during staffing changes. *Mitigation:* Maintain decision logs and backup contacts to prevent single points of failure.

R7: Withheld Critical Specifications. Security constraints may delay access to key platform data, increasing assumption-driven integration risk. *Mitigation:* Use phased NDA disclosure and formally record assumptions with customer sign-off.

5 Account Manager & Stakeholder Coordination

5.1 Account Manager Role and Responsibilities

Muhammad Ridwan Putra Jasni serves as Brigman's primary representative to Triton Deepwater, coordinating communications, clarification sessions, and escalation when required. This role provides consistent messaging and a single point of contact throughout delivery.

5.2 Stakeholder Engagement Strategy

Engagement used targeted sessions with operational stakeholders and visual models to align technical and operational perspectives. Decisions were documented, tracked, and validated through structured follow-up.

5.3 Escalation Protocol and Coordination Outcomes

Escalation follows four levels: technical resolution (24 hours), account manager mediation (4 hours), executive engagement, and contractual review when needed.

This approach resolved 86.7% of incongruities through client input and established clear coordination protocols for future phases.

6 Declarations

AI Tool Usage Declaration: ChatGPT assisted with language refinement and document flow. All technical analysis, models, and substantive content were produced by Brigman teams.

Originality Declaration: This report presents original analysis by Brigman Deep Sea Technology. BCIC is a proprietary framework applied specifically to this project.

Software Compliance Declaration: Analysis aligns with IEEE 830 and applies CRaM principles (Correct, Relevant, Measurable).

Software Scope Declaration: This analysis covers software requirements only. Hardware specifications and interfaces are customer-defined, and the software integrates those components.

Sign-off & Repository Information

BRIGMAN REPOSITORY INFORMATION

`https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113`

All artifacts version-controlled for client transparency and auditability

BRIGMAN TEAM APPROVAL SIGN-OFF

Brigman Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	
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Report Version: 3.0 (Full BCIC Methodology)
Client Delivery Date: February 12, 2026
Analysis Status: Complete - Ready for SRS Development
Client Next Step: Review and approve categorized requirements

Appendix A: Applied IRC Results

A-1 Complete Filled IRC Application Table

Table 5: Complete Filled IRC Application Table with All Project Findings

PS Ref	IRC Flags	Reason	Clarify?	Resolution & Action
BO-L1-4	CTX-1, CTX-2, CON-2	"Abnormal conditions" ambiguous and not testable; no definition	Y	Quantified: Pressure ±5%, temp ±10%, power ±15% thresholds
IT-L0-C2	IMP-1, CTX-2	Essential constraint with ambiguous language, no measurable parameters	Y	Specified: 1 kbps min bandwidth, ≤30s latency, 95% uptime
IT-L0-C3	CTX-1	"Protected in accordance with applicable security obligations" – undefined standards	Y	Specified: AES-256 encryption, X.509 certificates, audit logging
BO-L1-5	CTX-1, CON-1	"Orderly manner" ambiguous; compound continuation/termination/recovery	Y	Defined: Complete current operation, log shutdown reason, safe state transition
IT-L0-A1	CTX-1, CON-1	"Primary mission sensors supplied by customer" – missing interface protocols	Y	Specified: ROS2 middleware, DDS protocol, JSON format, API specs
BO-L0-A1	CTX-3	"Missions supervised by trained personnel" – out of scope for software	Y	Confirmed software-only scope; operational assumption noted
IT-L1-5	CON-1	"Operational Logging Capability" combines logging, storage, retrieval, analysis	N	Decomposed into 4 atomic requirements
BO-L1-6	IMP-3	"Operational Accountability" – missing clear responsibility assignment	Y	Revised: Software logging with operator ID attribution
BO-L2-4.1	IMP-2	"Abnormal operating conditions shall be identifiable" – not clear system obligation	N	Revised: "System shall identify abnormal conditions within detection thresholds"
BO-L2-4.2	CTX-1	"Safe operational state" – ambiguous safety condition definition	Y	Defined: Neutral buoyancy, minimal power, communication beacon active
BO-L1-4 (duplicate)	CON-2	References abnormal conditions without defining them	Y	Added specific definition with quantitative thresholds
BO-L1-5 (duplicate)	CON-1	"Mission Continuity and Recovery" combines continuation, termination, recovery	N	Decomposed into SYS-CONT-001, SYS-TERM-001, SYS-REC-001

A-2 IRC Application Quantitative Summary

Metric	Value	Percentage
Total PS Clauses Analyzed	27	100%
Total IRC Checks Applied	216	8 per clause
Incongruities Identified	15	55.6%
Imperative Issues Found (IMP series)	4	14.8% of clauses
Continuance Issues Found (CON series)	3	11.1% of clauses
Contextual Issues Found (CTX series)	8	29.6% of clauses
Requirements Requiring Clarification	13	86.7% of issues
Issues Resolved with Client Input	13	86.7% resolution rate
Requirements Ready for SRS	22	81.5% of original

Table 6: IRC Application Quantitative Summary and Performance Metrics

A-3 IRC Value Demonstration & Process Benefits

Benefit Area	Demonstrated Value & Outcome
Early Defect Detection	15 requirement issues identified before implementation, preventing potential rework costs estimated at $350K+
Structured Clarification	IRC framework guided efficient stakeholder discussions, reducing clarification time by approximately 40%
Quality Assurance	All requirements now meet CRaM principles (Correct, Relevant, Measurable) with clear verification methods
Regulated-Revision Guide	Systematic transformation of ambiguous requirements into precise, testable specifications with traceability
Risk Mitigation	Early ambiguity resolution reduced implementation schedule risk by 4-6 weeks and technical uncertainty
Client Confidence	Demonstrated rigorous, systematic approach to requirement quality and stakeholder collaboration

Table 7: IRC Framework Benefits and Project Value Demonstration

Appendix B: Risk Assessment Matrix Legend

B-1 Risk Rating Scale

The risk assessment matrix employs a systematic rating scale from 1 to 8 for both Impact and Likelihood dimensions. Table 8 provides the classification criteria used to evaluate each identified risk.

Legend	Impact	Likelihood
1	Catastrophic	Almost Certain
2	Critical	Very Common
3	Major	Highly Likely
4	Significant	Likely
5	Moderate	Moderately Likely
6	Minor	Somewhat Likely
7	Minimal	Unlikely
8	Negligible	Least Likely

Table 8: Risk Assessment Matrix Rating Legend

B-2 Risk Assessment Matrix

Impact \ Likelihood		Likelihood (L)							
		1	2	3	4	5	6	7	8
Impact (I)	1	1	2	3	4	5	6	7	8
	2	2	4	6	8	10	12	14	16
	3	3	6	9	12	15	18	21	24
	4	4	8	12	16	20	24	28	32
	5	5	10	15	20	25	30	35	40
	6	6	12	18	24	30	36	42	48
	7	7	14	21	28	35	42	49	56
	8	8	16	24	32	40	48	56	64

Table 9: Risk Assessment Matrix – Systematic Evaluation Framework